A Screening identity · reported contrast

Vaccinated mothers

Fraction of infants c

Risk $\propto 1 - \mathrm{VE}$

Unvaccinated mothers

Fraction of infants $1 - c$

Risk $\propto 1$

Incidence $\propto c(1 - \mathrm{VE}) + (1 - c) = 1 - c \cdot \mathrm{VE}$

Reduction from c_0 to c_1 $= 1 - (1 - c_1\mathrm{VE})/(1 - c_0\mathrm{VE})$

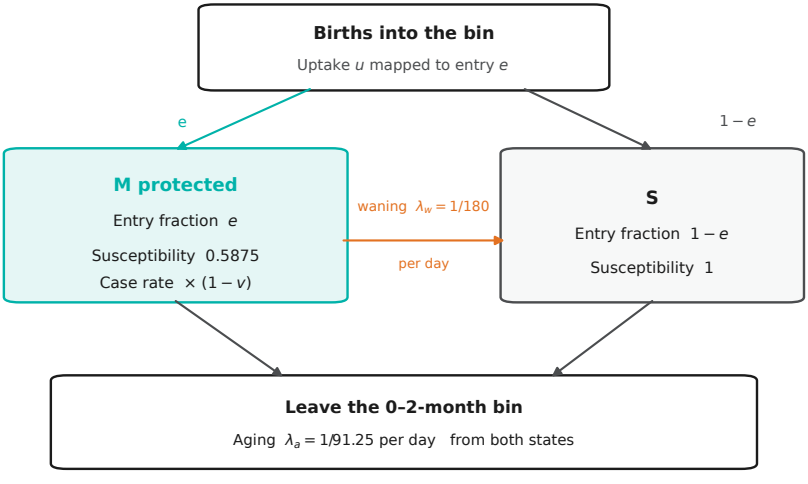
73.0% → 90% at VE 84.6% reduces incidence by 37.61%

Published VE limits 76.9–89.7% through the same expression: 29.80–44.18%

Onset before 2 months, VE 86.2%, same target to 90%: 39.53%

The vaccination odds ratio equals the disease risk ratio when c is the exposure prevalence. No slope, duration or transmission level enters.

B Transmission stack inside infant_0_2m · diagnostic, not the reported impact



0–2m	3–11m	1–4y	5–9y	10–17y	18–39y	40–64y	65+

Age bins. The highlighted bin is named 0–2 months and is 0.25 years wide.

0.57 and 0.02 are the line at uptake 1 and 0, not a measured conditional probability.

Boundary clearance is off, so protection can age forward. Within-bin equilibrium uses aging as the exit.

Closed form

$\varphi = 180/271.25 = 0.664$

Protected person-time $p = e\varphi$

Cases $\propto 1 - pv$

Legacy composite on that state

$v = 0.818$

Map endpoints 0.57 vs 0.02

Individual effectiveness 30.2%

31.2% if the composite is 0.846

90% target, direct formula

6.58% of bin cases

Integrated reduction 5.92%

0.66 pp left unassigned

Dose-1 share inside the bin is 0.

Screening contrast at the same target remains 37.61%.